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Energy optimization of the configurable service portfolio for IoT systems

Xiaojing Zhu[1]

School of Intelligent Science and Engineering, Yunnan Technology and Business University, Kunming, Yunnan 651701, China

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ABSTRACT

The development of the Internet of Things technology has promoted the continuous improvement of information technology in various fields such as industry, military and transportation. And communication and information exchange through the Internet, with the ability to sense, so smart devices came into being. First, this article proposes an IoT service framework that encapsulates each function of a smart device in an IoT environment into a service that is aggregated into a service class based on the similarity of its functions. Secondly, based on the particle swarm optimization algorithm, the correct service combination is recommended for the service chain. The gray correlation degree evaluation method is used to serve configurable resources, and the combined energy scheme is used for multi-objective comprehensive evaluation. Among them, the configurable service combination plan is evaluated from four aspects: service resource cost, planned completion time, overdue time and resource idle rate. The experimental results show that the particle swarm optimization algorithm can obtain a relatively good IoT service combination, which has better performance than genetic algorithms and ant colony algorithms, and can maintain the load balance of smart devices in load balancing, thereby effectively extending the entire network Life cycle.

1. Introduction

Before the practical application of the IoT, the research on energy optimization of service portfolio is mainly based on automated production equipment. The integration of production equipment information and optimization to complete the manufacturing task has limitations. In the configurable service combination capability optimization Manufacturing resources such as manpower and materials are also indispensable components for completing manufacturing tasks. With the development of intelligent manufacturing of the IoT [1–3], the creation of IoT technology in service portfolio energy optimization can be applied to dynamic resource objects. The information of people, manufacturing equipment, materials, products, services, etc. achieves accurate, real-time and reliable ubiquitous perception. On this basis, the dynamic energy optimization combination problem has also become a research hotspot.

With the continuous development of new generation information technology, the IoT [4,5] is regarded as another major change in information development after computers and the Internet. At present, in the face of the dynamic changes of the IoT environment, while mapping various resources and devices in the IoT environment into massive services, how to realize the dynamic combination of services to achieve optimal configuration is a major challenge. In response to the above challenges, many scholars at home and abroad have conducted in-depth research on the quality of service composition in the IoT environment, that is, mainly for different types of services and candidate services in the IoT environment, by aggregating candidate services under each category [6–8], to achieve a task-oriented overall QoS maximization. However, in the actual process, because the service information comes from the real-time data in the real world, the environment in which the service is located is dynamic and uncertain. Portable sensor system was constructed for detection of acetone vapor based on the cataluminescence(CTL) theory [9]. The inappropriate service combination is not only difficult to meet the user's demand for QoS in the IoT environment, but also Cause waste of service resources. Therefore, the optimization of the service portfolio quality assessment system under the IoT environment still needs further exploration and research. In addition, the overall energy consumption generated by the service composition in the IoT environment during the dynamic configuration process remains a key issue. Most of the current research focuses on the quality of service portfolio [10–12] in the IoT environment, and does not consider the energy consumption in the service portfolio optimization process. Therefore, in the QoS-oriented service portfolio optimization process, it is important to realize the configurable energy optimization of service composition quality in the IoT environment. The materials synthesis, apparatus and software, optimal conditions selectivity and stability were thoroughly studied [13]. However, the studies in the above literatures are mainly focused on the optimization of service composition based on QoS evaluation indicators, and do not consider the energy consumption in the process of service composition

E-mail address: ayako-midory@163.com.

[1] All author contributions are the same.

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optimization. Nor is there a separate assessment of the impact of energy consumption on mass services during dynamic provisioning.

As IoT technology enters industrial applications, coordination and collaboration between smart devices becomes an important strategy to meet the complex needs of users in industrial production [14–16]. The geometric features and electric conductivity of the power equipment storage scene affect the radio frequency propagation characteristics is discussed [17]. How to implement energy-aware service combinations in the IoT environment is also nowadays. One is of the research hotspots. In terms of functionality and computing power, a single smart device typically has limited capabilities, so a single smart device has only a limited number of functions. In this case, multiple smart devices can access the network and interconnect and collaborate through the network, which facilitates the completion of relatively complex functional requirements. In view of the above ideas, this paper proposes an IoT service framework in the IoT environment, which encapsulates each function of the intelligent device in the IoT environment into a service, and these services are aggregated into service classes according to the similarity of their functions. Second, recommend the right combination of services for the service chain. Considering the space–time constraints, residual energy and functional constraints of IoT services on the same smart device, the IoT service is reduced to a multi-objective optimization problem and heuristic algorithm is used. Solve. Finally, a comparative experiment is carried out from three aspects: the recommendation result of the service combination, the minimum residual energy of the smart device and the load balancing of the entire network. The experimental results show that in the context of this paper, the particle swarm optimization algorithm can obtain a relatively better IoT service combination, and its performance is better than genetic algorithm and ant colony algorithm, which can maintain the load balance of smart devices in the network, thus effectively extending the whole the life cycle of the network.

The rest of this paper is organized as follows. Section 2 discusses Service portfolio energy platform for the IoT, Section 3 introduced Configurable service portfolio energy optimization, Section 4 shows the simulation experimental results, and Section 5 concludes the paper with summary and future research directions.

2. Service portfolio energy platform for the IoT

The increasingly broad evolution of the IoT concept raises higher and more complex requirements for both the network as an infrastructure and the technology for communicating between heterogeneous intelligent objects through the network. In an IoT environment, the data (or events) generated by a large number of IoT devices and applications are presented to the user as a final result, or as input to other applications for consumption by other applications. The same data may be needed by multiple users or multiple applications [18–21]. Many efforts have been devoted to synthesize decorated materials with excellent CTL performance [22,23]. However, unlike traditional applications, in the IoT environment, data is not pulled by the consumer, but is pushed by the system or middleware to the data consumer after the data producer publishes the data. The event-driven mechanism is in line with the characteristics of the IoT-aware service and is the basis for implementing intelligent and active services.

As an influential future network architecture system, SDN (Software Defined Network) [24] introduces the concept of centralized control. Under the open architecture of SDN, a certain range of networks (also called an SDN domain) can be managed by one or more logically centralized control units. In this way, the design, deployment, maintenance, and management of a domain can be completed in the control unit, thereby eliminating the differences in the underlying network. This logical control unit is the SDN controller responsible for running the network operating system and controlling all network nodes.

2.1. IoT objects, IoT services

IoT devices refer to ubiquitous end devices and facilities, including sensors, actuators, Bluetooth, wireless sensor networks, embedded systems, and more. Their development is mainly driven by the development of technologies such as Micro-Electro-Mechanical (MEMS) technology, sensor technology, radio frequency identification technology and Near Field Communication(NFC). IoT devices can automatically record data, process information, and send it to the network.

At present, the widely recognized IoT architecture is a three-layer structure including an application layer, a network layer and a perception layer. We proposed service portfolio optimization framework based on IoT, its architecture definition is shown in Fig. 1.

In Fig. 1, the perception layer acts like a nerve ending, and its main function is to identify physical entities and collect information. The network layer realizes information transmission through various existing wired networks and wireless networks. Its role is like the nerve center and the brain, used to transmit and control the data obtained from the sensory layer. The application layer is responsible for interacting with users to implement smart applications.

2.2. WEB service in the IoT environment

As the product of the expansion and extension of the Internet to the physical world, the IoT realizes the connection and interaction between the real world and virtual information systems. In order to achieve communication, information sharing and exchange between various physical entities to realize the identification and management of these physical entities, it is necessary to build an IoT software architecture, call Web services in different environments, and have a single function Web. Services are integrated into a coherent, full-featured application.

Web Services Description Language(WSDL) is a standard service description specification built on top of the message layer and is currently the most widely used. WSDL is an XML-formatted description document that describes the functional information of a Web service, including data information, interface information, and binding information. Web services in WSDL represent a set of ports that can be accessed by users, describing where and how to invoke a service. It describes the access address of the service when describing a service. All operations of the service, such as the parameter information that must be carried and the return value information after the operation is executed, and the binding information of the specific transport protocol.

2.3. Configurable service queue combination

The Open Flow protocol specifies that the number of outbound port sub queues can be set, but at most eight sub queues are allowed. In this paper, according to the granularity of the differentiated service, 3 sub-queues are set for each outgoing port. If the sub queue is set too much, the complexity of queue management will be greatly increased. The bandwidth of the output port is allocated to each subqueue as the minimum data rate for each subqueue. In addition, the minimum data rate for each subqueue is not static, and SDN-C can be modified according to certain policies.

As Fig. 2 is shown, for queue bandwidth adjustment, we set three states for the queue: good, light congestion, and heavy congestion. Since queue 1 has the highest priority and the lowest packet loss rate requirement, the queue management mechanism gives queue 1 a priority bandwidth guarantee. When SDN-C detects that queue 1 is in a lightly congested state, it should other queues ‘borrow’ bandwidth to meet the latency requirements of queue 1. Other queues should unconditionally ‘borrow’ some or all of their bandwidth.

In a traditional network, each routing or switching device is responsible for selecting routing paths and discarding packets that cannot be delivered. Although each packet may have different QoS(Quality of Service) restrictions, the ‘best efforts’ service provided by the routers on

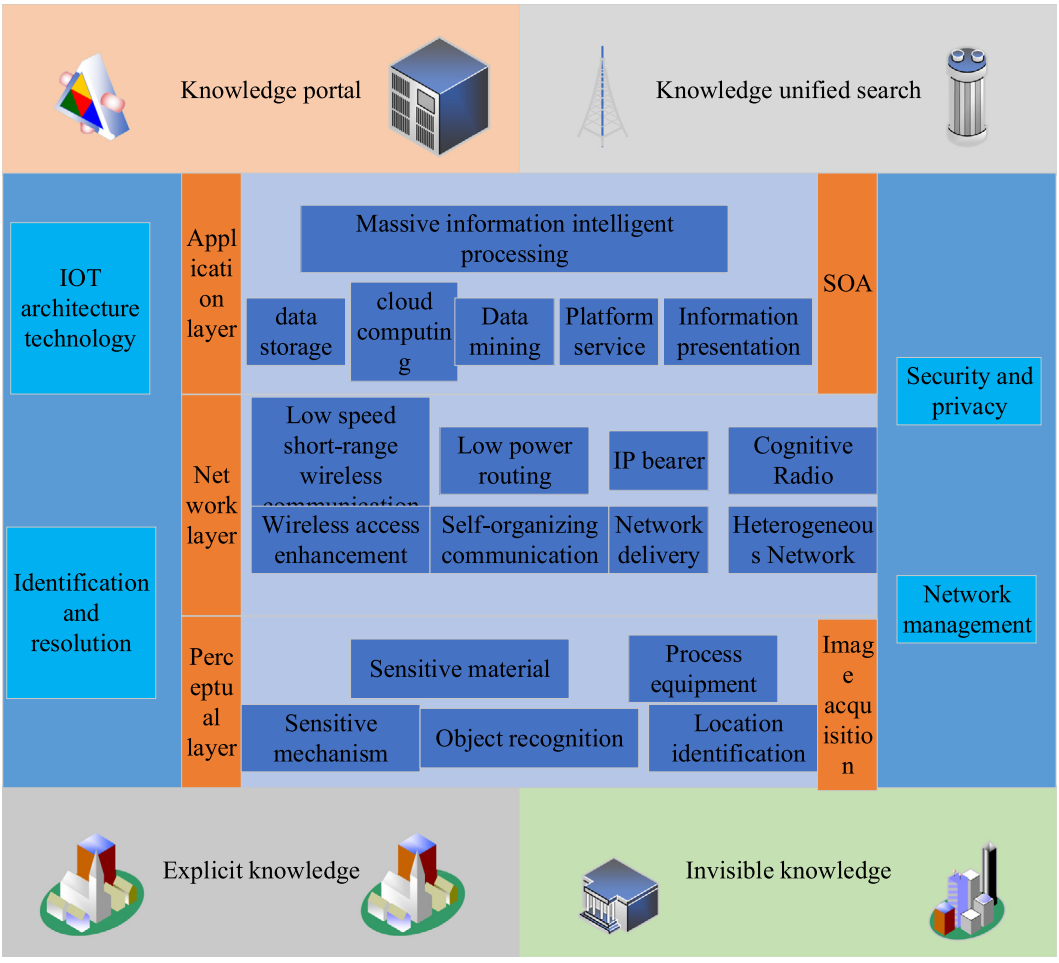


Fig. 1. Service portfolio optimization framework diagram under the IoT.

the traditional network cannot provide differentiated services according to the QoS requirements of the per-packet, and thus cannot meet the diverse transmission requirements of the IoT. The subscribe system adopts SDN as the basic network facility. Considering the characteristics of the SDN network, SDN-C first calculates the routing path based on global topology knowledge and network state, and then installs the routing path on the switch on the path. The event flow (by flow) has calculated the routing table when SDN-C enters the network. However, only the routing forwarding path to each stream is insufficient. According to the queue management scheme proposed above, each forwarding of the switch each priority queue is set up on the egress. Each priority queue guarantees that its queuing delay is bounded. Therefore, the priority queue that per-flow enters at the egress port of each switch also affects the total delay of the flow. That is to say, the scheduling of per-flow on each switch is still important to ensure its QoS requirements.

$$T_i = T_{\mathrm{i}}^{\mathrm{Proc}} + T_{\mathrm{i}}^{Que} + T_t^{Tran} \tag{1}$$

T_{i}^{Que} represent the queuing delay, the delay from the time the P enters the sub-queue of the egress port to the time it is forwarded, and the delay is related to the queue length and the bandwidth allocated to the sub-queue. T_t^{Tran} indicates the transmission delay, which is the time required to put all the bits of packet p on the link, and its value is related to the bandwidth of the link and the length of P; $T_{\mathrm{i}}^{\mathrm{Prop}}$ stands for propagation delay, which refers to the time required for an electrical or optical signal to travel on the link, and is related to the medium used for transmission and the transmission distance; $T_{\mathrm{i}}^{\mathrm{Proc}}$ stands for processing delay, which refers to the time required for the switch to parse packet header information, query routes, and perform necessary re-encapsulation. For wire-speed switches, the time is usually considered constant and measurable, and the magnitude is usually micro.

Assuming all switches are configured the same, $T_{\mathrm{i}}^{\mathrm{Proc}}, T_{\mathrm{i}}^{\mathrm{Prop}}$ is considered constant and measurable. Then, for a path with m relay nodes, the total delay is:

$$P^i = \sum_{i=1}^{n} (T_i^{Que} + T_i^{Tran} + T_i^{\mathrm{Proc}}) + (m+1) * T^{\mathrm{Prop}} \tag{2}$$

Thus, the queuing delay is actually the only non-deterministic factor in the delay sum. Other delay elements can be measured by experiments similar to the literature. Propagation delay and processing delay can be obtained only once in the case of low load: a small topology with only one controller and two SDN switches is set up, and a time-stamped measurement packet is sent from the controller, after passing through a switch. The controller is re-delivered, and the controller calculates the time difference between sending and receiving the packet, which is represented by T_1, and then:

$$T_1 = 2 * T^{prop} + T^{\mathrm{Proc}} \tag{3}$$

Then, based on the above topology, add a switch, that is, two switches in series. In the same way, only the test packet is forwarded by two switches, and finally reaches the controller. The time difference is recorded as T2, then T2 has the following structure:

$$T_2 = 3 * T^{prop} + 2 * T^{\mathrm{Proc}} \tag{4}$$

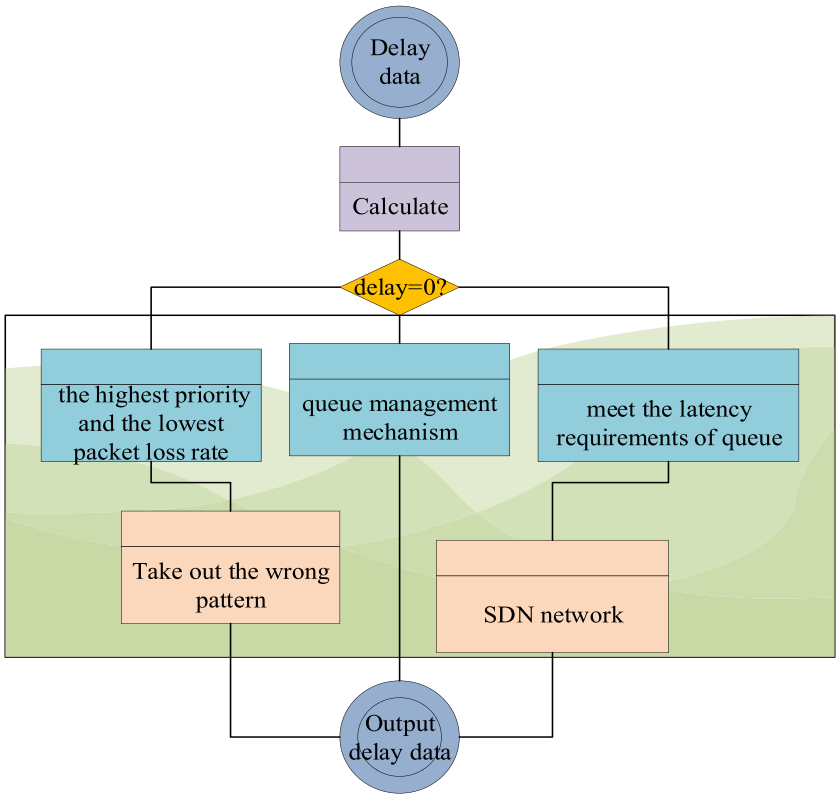


Fig. 2. The queue bandwidth adjustment algorithm.

Table 1
Sub-queue genetic mapping algorithm pseudo-code description.

Input: route multicast tree for stream f Output: sub queue selection on each path
1: Initialize the output matrix O with inf, O is used to record the queue parameters; 2: traverse the forwarding tree to get all paths from root to leaf, represented by path set P; 3: Sort the paths in the path set P by the length of the path from long to short, and the result is still stored in P; 4: If all the paths in P cannot satisfy the delay limit {C} of the stream f, the program is exited; 5: Continue with the following steps for each path in P; 6: Query the matrix O, and obtain the attribute value of the node on the path that has been allocated the sub-queue by the already processed path; 7: taking into account the above results, calculate the most suitable sub queue sequence on the path; 8: If there is a sub queue sequence that satisfies the delay requirement C_f, the result matrix O is modified; 9: According to the result matrix O, reconfigure the flow entries on the relevant switch.

The routing of configurable service queue events is calculated based on the subject. Since each topic may have multiple recipients. Therefore, the route uses multicast routing. Its routing path is a forwarding tree with the sending source as the root node. The path from the sender to each receiver needs to meet the latency requirements of the stream. Since the propagation delay is only a small part of the delay configuration of each path, it is reasonable to assume that the longer the path, the larger the delay.

The multiple paths of each stream are sorted according to the length of the path, and the sub-queues are preferentially assigned to the path with the longest path. The other queues then allocate sub-queues for other router egress ports based on the shared path that has completed the queue mapping. The algorithm is shown in Table 1.

Line 3 is sorted using direct insertion. Line 4 first checks if the estimated delay when all the highest priority queues are used on the shortest path meets the delay limit C. If it is not satisfied, the network cannot meet the QoS requirement of the flow f, and the flow needs to be discarded and the program is exited. Lines 5–8 select the most appropriate sub queue sequence for each path in order of length to short. If there is no sub queue sequence on the path that meets the delay limit, skip the path and continue calculating. The sub queue sequence of the next path; otherwise, the result matrix O is modified with the result of the calculation. Finally, line 9 reconfigures the flow table for the associated switch.

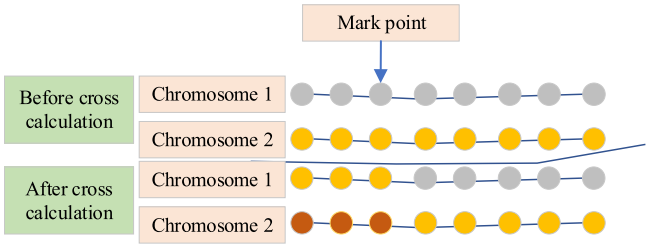


Fig. 3. Chromosome crossover process.

2.4. Service composition energy optimization algorithm

(1) The crossover operator plays a major role in genetic algorithms. Crossover refers to the process of replacing part of the structure of two parent individuals to produce two new individuals. In the new generation population, two chromosomal individuals are randomly selected, and the markers are made at the same positions of the two selected chromosomal individuals, and then the chromosomal sub-segments before or after the marked position are exchanged to generate new chromosomal individuals. Fig. 3 shows the process of crossover between two chromosomal individuals.

(2) The path of the ant colony constitutes the solution space of the problem. According to the different paths chosen by different ants, ants with shorter paths will release more pheromones. In the long run, the concentration of pheromone accumulated on a shorter path becomes higher and higher, and the ant chooses the direction of the path according to the pheromone content, and the number of ants on the path will also increase. The entire ant colony will focus on the optimal path driven by the pheromone concentration. This is the optimal solution to the problem. The algorithm mainly includes two steps of establishing the initial path and updating the pheromone matrix.

(3) In the particle swarm algorithm, the feasible solution of each optimization problem is a bird in the search space, which is called "particle". Each particle has a memory function, which can remember the best position searched, and there is a speed that determines the distance and direction that these particles fly. Particle swarm optimization is the process by which particles in space search for the optimal solution in the solution space following the current optimal particle.

3. Configurable service portfolio energy optimization

The IoT itself has some characteristics different from the Internet. The IoT is the expansion and extension of the Internet to the physical world. Therefore, the characteristics of IoT services will be affected by physical equipment and application fields. This will lead to a certain difference between the Web service composition in the IoT environment and the traditional Web service combination. The starting point of the traditional Web service composition is the user's functional requirements for the service, and sometimes non-functional requirements, such as price and response time. Wait. However, the Web service composition in the IoT environment cannot simply integrate the Web service that meets the user's requirements into a new service based on the user's requirements. It needs to be based on the user's needs, but the trigger of the combined service execution. The conditions are largely due to events from physical devices.

The main task of process definition management is to manage process definitions, including the design and deployment of process definitions. The main tasks include:

1. Analyze user is requirements, abstract the logical relationship of the user request and the atomic services needed to complete the service according to the service description submitted by the user.

2. Business process definition, First query the business process definition library, if there is already a process definition that meets the user's requirements, instantiate the process definition and execute; if the required process definition is not met, then combine the QoS management component to the appropriate Web service from Universal Description Discovery Integration (UDDI), based on the service Query between business process modeling and process definition files.

3. Service query and binding, combined with the QoS management component, query the service list of the UDDI registry, find the appropriate Web service, form the corresponding combined service, design and generate a process model that meets the user function charging function requirements, and convert Define a file for the process and store it in the business process library.

4. Process-defined deployment, deploy the defined business process, and save it to the business process library. In the object-oriented web service environment, in order to meet the various needs of users, combined services have become an inevitable trend. However, due to the unknown nature of the environment and the arbitrariness of services, the quality of web services is also highly uncertain. With the development of composite services, the quality of research has also been significantly improved.

Wireless sensor networks include a diverse distribution of sensors and actuators to represent physical characteristics or execute commands. The demand for applications determines the diversity of QoS requirements for wireless sensor networks. At present, the research on QoS of WSN includes SAR (Sequential Assignment Routing), minimum consumption forwarding, SPEED, Energy Aware Routing, MMSPEED (Multi-Path Multi-Speed Protocol), DAST. In this study, the WSN's QoS aware routing protocol is proposed to provide appropriate QoS and bandwidth limitations for energy in the personal environment.

After analyzing the QoS metrics of the composite service and the WSN, it can be seen that their requirements are inconsistent. Among the five QoS metrics, the price is set in advance by the service provider, and the reputation is set by the service user according to the experience of use, which is a subjective indicator. The relationship between QoS objective metrics and metrics is described below.

The QoS metrics of the composite service need to be decomposed, mapped to the WSN, and used to obtain the overall quality of the service. The overall QoS is limited by local QoS and needs to be obtained through the QOS management module. In addition, composite services in the IoT also use QoS metrics from other Web services, which are provided by other functional entities.

The composite service is described in the form of a typical workflow diagram. The form of composite services can be divided into serial mode, parallel mode, branch mode and loop mode.

$$\begin{cases} G = \{V, E, G\} \\ V = (S_i) \\ E \subseteq V * U \\ Q = \{q_1, q_2 \cdots q_n\} \end{cases} \tag{5}$$

G is a directed graph of the WSN. In V, S_i is the computed set of WSNs, including a single sensor and the network to which it is connected. E is the relationship between S_i. Q refers to the QoS metric. G is the response time in milliseconds. Gr is a credibility indicator. The value range is [0, 1]. The value of the indicator is [0, 1]. The value is determined by the specific indicator and transmission of the sensing network.

$$\begin{cases} q_r(s) = \sum_{i=1}^{n} q_r(s_i) \\ q_r(s) = \prod_{i=1}^{n} q_r(s_i) \\ q_a(s) = \prod_{i=1}^{n} q_a(s_i) \end{cases} \tag{6}$$

In parallel mode, the system requires the device service S_i to take the same amount of time to perform the task. For example, motor vehicle management systems, detecting GPs sensors, fuel sensors, temperature sensors, tire pressure sensors. When the information is collected, a detailed report on the results will be sent to the middleware. When any of the sensors are not working, the task is not completed.

In the first model, if the sensor is limited by energy, then it needs to extend the WSN usage time as much as possible. We assume that the probability of each branch execution is p_t, which is defined according to user requirements. For example, if the WSN is set according to the minimum time optimization, then the probability of having the lowest delay is set to 1, and the probability of the remaining branches is set to 0. If the energy of this branch is exhausted, then the lowest delay probability of the branch Is 1. So we can set the value according to the loop method.

$$\begin{cases} \sum_{i=1}^{n} p_t = 1 \\ q_t(s) = \sum_{i=1}^{n} p_t q_r(s_i) \\ q_a(s) = \sum_{i=1}^{n} p_t q_a(s_i) \end{cases} \tag{7}$$

The second model gives priority to usability and credibility indicators. When the first trial condition is completed, the task is declared complete and reported to the system.

$$\begin{aligned} q_r(s) &= \min_n \left\{ q_r(s_i) \right\} \\ q_r(s) &= 1 - \prod_{i=1}^{n} (1 - q_r(s)) \\ q_a(s) &= 1 - \prod_{i=1}^{n} (1 - q_a(s)) \end{aligned} \tag{8}$$

S_i runs in sequence, and each cycle time is set according to the cycle time L.

$$\begin{aligned} q_r(s) &= L \left\{ q_r(s_i) \right\} \\ q_r(s) &= (\prod_{i=1}^{n} (1 - q_r(s)))^L \\ q_a(s) &= \prod_{i=1}^{n} (1 - q_a(s))^L \end{aligned} \tag{9}$$

In the workflow engine, the operation of the web service, the service location information required to access the operation, the parameter information, and the like need to be described, wherein the description of the parameter includes the name, the type, and whether the parameter is an input parameter or an operation required to access the operation. The result is of the execution. In order to facilitate the workflow engine to make a reasonable and orderly call to the web services bound to each active node based on this information.

In order to facilitate communication and computer understanding, the following standardized descriptions of Web services are performed using XML. After the engine is started, the engine initialization module first performs an "initialization engine" activity to initialize the engine configuration based on the engine configuration file information. After the engine is initialized, the Parse Process Definition File activity adapts the process definition file parsing to the process model and stores it in the database. In the execution phase, the engine first creates and starts a process instance, the process executes to the active node, and the active node binds the external application. When an external application activity is executed, the corresponding external application is called, and the result data returned after the call is called and stored in the process variable. The process then continues until the process ends and the process execution record is saved at the end of the process.

According to the candidate service set formed by the workflow diagram, different combinations of combinations can be formed by arrangement and combination. The system needs to select the optimal combination scheme. This study uses the gray correlation degree evaluation method to serve the configurable resources. The combined energy scheme is used for multi-objective comprehensive evaluation. The configurable service portfolio scheme is evaluated from four aspects: service resource cost, planned completion time, overdue time, and resource idle rate. The calculation method of the combined scheme is as follows:

(1) The service cost C considered in this study is mainly composed of the labor cost C_W of the resource service portfolio, the equipment depreciation cost C_e, and the material scrap cost C_m, 3 part.

(2) Planned completion time T refers to the time when the current real-time status of the resource service combination is expected to complete the manufacturing task. The reference value is the delivery date of the manufacturing task $T_P = \max \left\{ T_W, T_e \right\}$. Among them, T_W indicates the completion time of the worker plan, and T_e indicates the manufacturing equipment plan.

(3) Overdue time means the completion of the manufacturing resource service plan the time of the work exceeds the delivery time required by the manufacturing task. Overdue will have a great impact on the production schedule of the task, so it is an important time indicator $T_{ot} = \max \left\{ 0, T_P - T_l \right\}$. Among them, T_l represents the time required for the delivery date.

(4) Resource idle rate S_f refers to the idle situation of the manufacturing workshop under the demand of manufacturing tasks. As a manager, the waste of resources is not tolerated. $S_f = \frac{s_R^i}{s_N^i} (i = 1, 2, 3)$, 1 is human resources, 2 is material resources, 3 is Device resources. s_R^i indicates idle resources, and s_N^i indicates the total number of resources of this type.

In the following, the evaluation method based on the gray correlation degree is used to evaluate the combination scheme, and the optimal scheme of the manufacturing resource service combination is obtained. The relevant steps are as follows:

Step 1 Establish the original evaluation indicator matrix:

$$S = \begin{bmatrix} S_1^1 \dots S_n^1 \\ \dots \\ S_1^m \dots S_n^m \end{bmatrix}_{m \times n} \tag{10}$$

where m represents the number of manufacturing resource services to be evaluated in the manufacturing resource service candidate set; n represents the number of evaluation indicators, where n = 4. Step 2 determines the optimal indicator sequence, based on the indicator values of each alternative and manufacturing The demand of the task determines the optimal index. The minimum value of the cost type indicator is the optimal index, and the maximum value of the benefit type indicator is the optimal index, forming the optimal index sequence.

$$S^* = (S_1^*, S_2^* \dots, S_n^*) \tag{11}$$

Step 3: Initialization standardization

$$\begin{aligned} S_j^+ &= \max_{1 \le i \le m} \left\{ S_j^i \right\} \\ S_j^- &= \min_{1 \le i \le m} \left\{ S_j^i \right\} \end{aligned} \tag{12}$$

Step 4: Calculate the correlation coefficient

$$E = \begin{bmatrix} \xi_1(1) \dots \xi_1(n) \\ \xi_2(1) \dots \xi_2(n) \\ \dots \\ \xi_m(1) \dots \xi_m(n) \end{bmatrix}_{m \times n} \tag{13}$$

Step 5: Set evaluation indicator weights

According to the actual situation of the manufacturing task, the scene is different, To customize the weight of each evaluation index, the more mature method has AHP analytic method, and the weight is determined by comparison between the two indicators. The weight sequence set here is $u = (\omega_1, \omega_2 \dots \omega_n)^T$.

Step 6: Comprehensive evaluation results

The relationship degree of the manufacturing resource service plan is R (S), then $R(S) = Eu$. The comprehensive evaluation result is: Let $R^i = R(S^i)$, the greater the correlation degree R' value, the higher the comprehensive evaluation index value of the service portfolio candidate solution set i, and the better the configurable resource service combination scheme can better complete the configurable task.

4. Experimental examples

The entire system is carried out on a simple SDN test bed. The test bed is a topology based on virtual plus physics implemented by Open Switch and KVM. Among them, the hardware is composed of three IBM System x3850 XS servers running Server 12.04 operating system; Open switch adopts version 2.3.0 in software environment, Open flow adopts version 1.3, KVM uses qemu-kvm-1.0 (virtual machine management) Adopt version 1.3.0.

The experiment sets all link data rates to be 1O Mbps. The test packet length is 200 bytes. It should be noted that the use of virtual technology can impair the performance of the system and can give the system a more conservative evaluation. The performance of the system is evaluated from the following aspects.

(1) Delay and loss rate in one-to-one transmission

This set of experiments was used to study the end-to-end latency of the 1:1 transmission mode (one publisher and one subscriber). This section tests the delay and data for a 1-hop path (from node 1 to node

Table 2
Average delay and packet loss rate in transmission mode.

Hop count	Source node	Destination node	Average delay (ms)	Average packet loss rate
1	1	2	2543.67	4.31
2	1	4	2791.07	3.29%
3	1	6	15 908.67	0.00%

Table 3
M average delay and packet loss rate in m transmission mode.

Hop count	Source node	Destination node	Average delay (ms)	Average packet loss rate
1	1	2,3	2585.67	5.85%
2	1	2,4,5	13 593.07	0.01%
3	1	2,4,6,7	18 777.67	0.00%

Table 4
Average delay and packet loss rate in m:1 transmission mode.

Hop count	Average delay (ms)	Average packet loss rate
1	4872.0	6.41
2	22 138.0	0.12%
3	23 843.0	0.00%

2), a 2-hop path (from node 1 to node 4), and a 3-hop path (from node 1 to node 6). In each case, 100,000 UDP packets are continuously sent from the source to the destination, the first packet is sent as its start time, the last packet is received, and the end time is repeated, and the result is recorded 10 times. Table 2 is a record of the experimental results.

As can be seen from Table 2, the average delay increases rapidly is as the number of hops increases. However, the experimental results have an interesting phenomenon: the more hops, the greater the delay, but the packet loss rate is reduced, mainly due to the performance of the virtual machine used in the experiment. Sending a large number of packets continuously by the sender will cause the receiver to drop the packet because it is too late to receive. The limited reception capacity of the virtual machine limits the performance of the system.

This set of experiments was used to study the end-to-end latency of the 1 to m transmission mode (one publisher and multiple subscribers). The delay and packet loss rate are in the case of 1:2. 1:3 and 1:4 transmission modes and mixed hops were tested separately. Table 3 records the experimental results.

Compared with Table 3, the increase in the number of subscription nodes does not have a large impact on the delay (the results of the three sets of comparison experiments are relatively small). If a one-to-many transmission uses unicast transmission, the number of packets that flood into the network will multiply, which will greatly impair the efficiency of the system. Therefore, in the case of one-to-many transmission, the system should use multicast.

A set of experiments was performed in the same way to verify performance in a many-to-one transmission mode. Each group of experiments uses two senders to simultaneously send 100,000 packets to a subscriber. The number of hops required between the publishing node and the subscribing node is as shown in Table 4. The results show that the delay increases with the increase of the number of hops, but the packet loss rate decreases. This also shows that the bottleneck is still at the terminal node.

Based on the same experimental conditions, the effect of the number of flow entries on delay and packet loss rate was tested. The average delay and packet loss rate in the case of different hop counts and different flow table entries in the one-to-one transmission mode were tested. Experiments show that with the increase of flow table projects, the average delay is on the rise, but the change is very small. However, changes in the flow table project have no significant effect on the average packet loss rate.

Table 5
Effect of the number of flow entries on the average delay and packet loss rate.

Hop count	Flow entry	Average delay (ms)	Average packet loss rate
1	1000	2476	4.78%
	5000	2927	2.15%
	10 000	3044	4.98%
	50 000	3200	2.53%
2	1000	2791	3.29%
	5000	3325	0.00%
	50 000	3574	0.21
3	1000	15 098	0.00%
	5000	17 486	0.00%
	50 000	17 959	0.00%

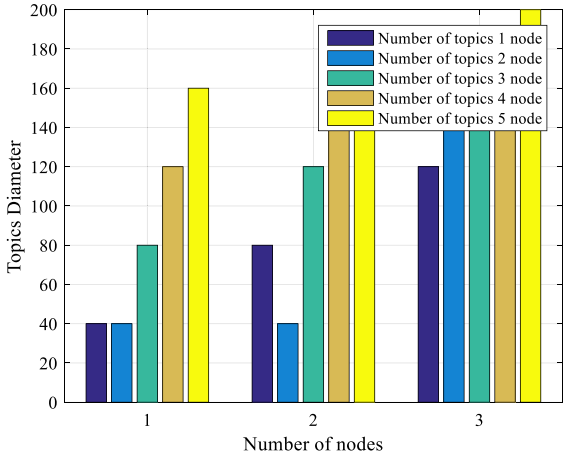


Fig. 4. Algorithm execution time in three cases.

As can be seen from Table 5, the lower the delay means the higher the packet loss rate. As mentioned earlier, the processing limitations of virtual machines have largely contributed to this phenomenon. This lab mainly evaluates the ability of switches and terminal devices to handle large numbers of events in a single hop situation. A publisher sends packets at different rates. After a certain rate is exceeded, the terminal node will not be able to discard the received packets in the future. Repeat this experiment several times and discover the fact that when the publisher sends 10,000 packets and the delay is 3044 ms, the packet loss rate is quite low. In another set of comparative experiments, using a PC instead of a virtual machine, in order to maintain the same low packet loss rate, the average delay of transmitting 500,00 packets must be maintained 17 959 ms.

The number of topics on the topic tree set in the experiment is different [40, 80, 120, 160, 200]. The number of nodes on the network (including the publishing node and the subscribing node) is 3, and these nodes are randomly generated. Half of the topics in the theme are on the first level of the theme tree. The experimental results are shown in Fig. 4. Note that the time amplitude is 180 ms. When the topic tree is a single tree, the execution time is the shortest. Otherwise, there is no coverage relationship between the topics, and the algorithm will take the longest execution time.

The parameters in Fig. 5 are selected as follows, D is fixed at 72, and C is changed from 100 to 1000. The experimental results show that the fitness function of the MFO under the QoS evaluation model is better than PSO and DE. In Fig. 5, C is fixed at 100, and D changes from 72 to 720. In Fig. 5, the fitness function under the QoS evaluation model obtained by MFO has obvious advantages over PSO and DE. Particle Swarm Optimization (PSO). Each particle in the particle swarm represents a possible solution to a problem. Through the simple behavior of individual particles and the interaction of information within the group, the intelligence of the problem is achieved. Differential Evolution Algorithm (DE) is an efficient global optimization algorithm.

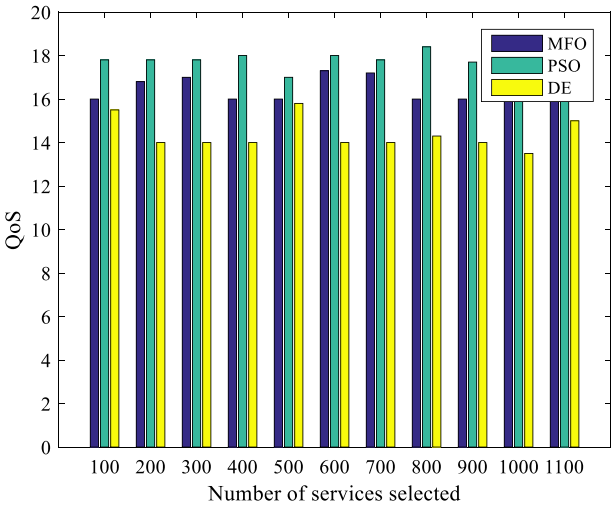


Fig. 5. QoS change curve of the number of categories.

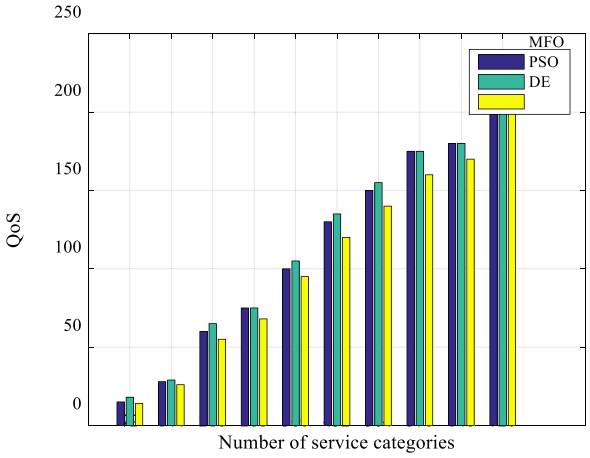


Fig. 6. Changes in QoS with the number of service classes.

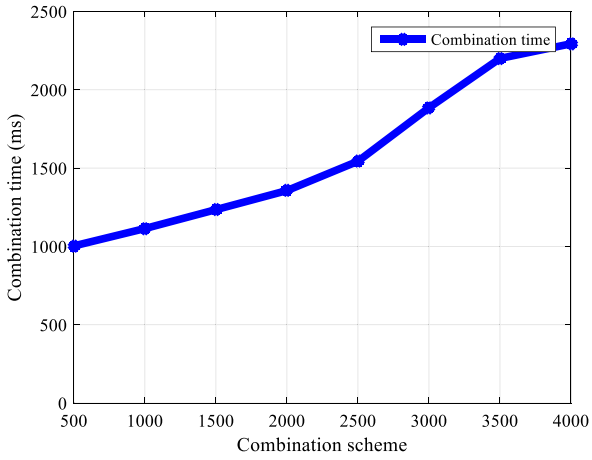


Fig. 7. Combination time from a combination of 500 to 4000.

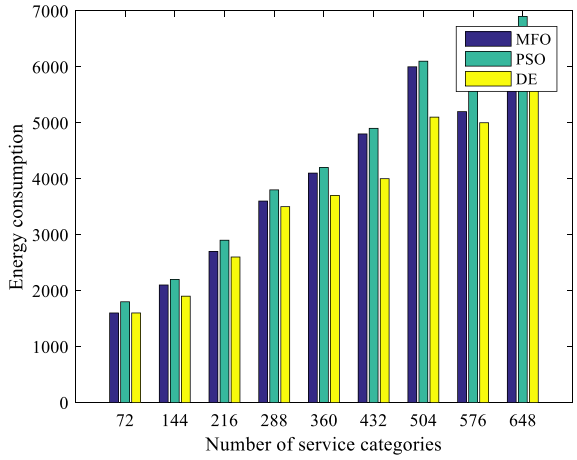


Fig. 8. Energy consumption varies with the number of service categories.

It is also a group-based heuristic search algorithm. Each individual in the group corresponds to a solution vector. At the same time, the fitness function value of the QoS evaluation model in Fig. 5 increases with the number of service categories, which can be explained as a linear relationship between the fitness function and the number of service categories (see Fig. 6).

In order to test the configurable resource combination scheme from 500 to 4000 combinations, repeat 8 experiments at a time (Fig. 7).

As shown in Fig. 7, when the combination scheme reaches 4000, the combined time consumed can still be kept within 2500 ms.

In order to illustrate that the selected algorithm obtains the best QoS, it does not affect the energy consumption of the service combination. Therefore, the parameter selection in the energy consumption assessment model is consistent with the parameters in the QoS model. This paper selects MFO, PSO and DE to evaluate the energy consumption of the service portfolio under the IoT service environment. In Fig. 7, D changes from 72 to 648, and C is fixed at 100. The experimental results show that under the above circumstances, the energy consumption of the MFO in the IoT environment and the energy consumption of the DE algorithm are significantly less than the PSO.

In addition, Fig. 8 shows that as the number of category services increases, the energy consumption also increases. This is because the energy consumption is generated at the same time as each service category is selected, and the total energy consumption and the number of service categories generated during the service selection process are presented. a linear relationship.

In addition, the execution success rate of the service combination when using the active discovery mechanism and not using the active discovery mechanism is compared (Fig. 8).

As shown in Fig. 9, since the module for real-time updating context-aware information is added to the manufacturing resource service ontology in the active discovery mechanism, the manufacturing resource service is more effective in matching the requirements of the manufacturing task, thereby improving the success rate of the combination. The rate is still reduced as the number of manufacturing tasks increases, but the use of context-aware based active discovery mechanisms can keep the combined success rate at a relatively high level.

In addition, the performance of the algorithm is compared by using the genetic algorithm, ant colony algorithm, and particle swarm algorithm's objective function value changes, the minimum remaining energy change, and the variance of the remaining energy of the entire network, as shown in Figs. 10, 11, and 12, respectively.

Fig. 10 shows the change trend of the objective function values of genetic algorithm (GA, ant colony algorithm (ACO) and particle swarm algorithm (PSO) after 30 consecutive executions. These objective function values are a standard for measuring whether the IoT service portfolio is relatively optimized. As can be seen in Fig. 9, the objective function value of PSO is relatively stable, approximately around −0.6, which is approximately a stable straight line, while the

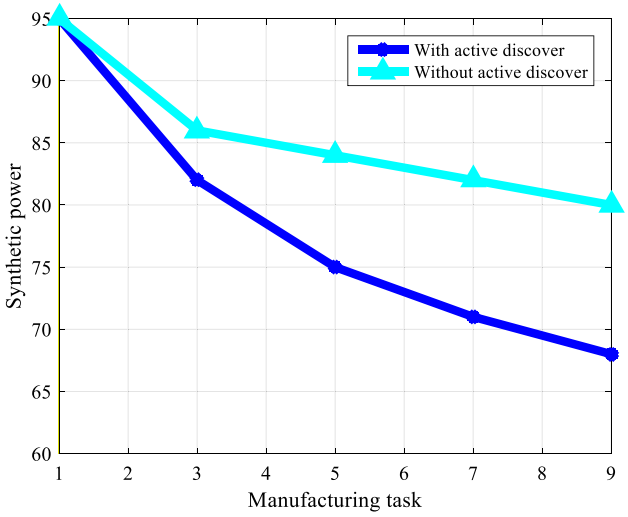


Fig. 9. Group synthesis power.

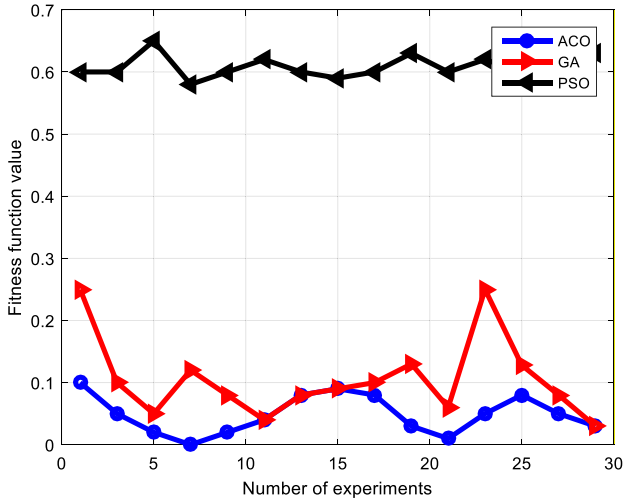


Fig. 10. Changes in the objective function values of the GA, ACO and PSO algorithms for 30 consecutive executions.

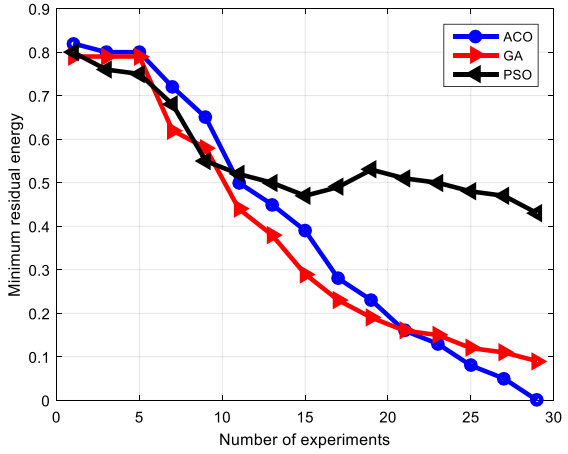


Fig. 11. Changes in the minimum residual energy of a smart device after 30 consecutive executions.

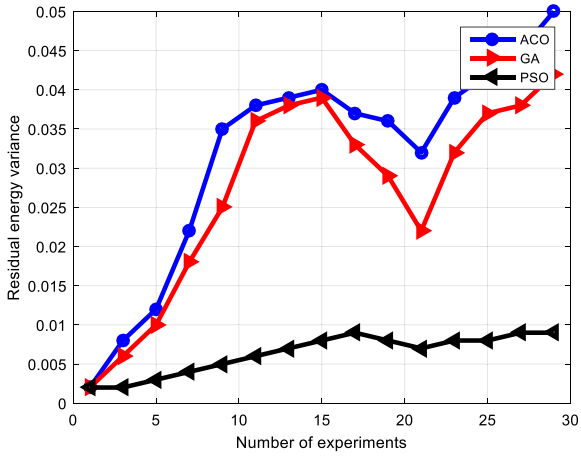


Fig. 12. Variation of residual energy variance of a smart device after 30 consecutive executions.

objective function values of GA and ACO performed continuously for 30 times fluctuate greatly, but generally show Slight upward trend. Experimental results show that under the current parameter settings, the PSO algorithm is more likely to generate a relatively better service combination in most cases, which means that PSO is more likely to generate a relatively optimal solution under the current conditions. With the increase of the number of iterations, the energy consumption of smart devices is getting larger and larger. The ratio of instantiating a service to the remaining energy of the smart device becomes larger, and the load balancing factor also increases, which leads to the goal. Increase in function value.

In order to evaluate the effectiveness of the algorithm, Fig. 11 show the change of the minimum residual energy of the intelligent device in the network obtained by performing 30 consecutive executions. As can be seen from the figure, compared with the minimum remaining energy of GA and ACO, which decreases rapidly, the PSO algorithm is used to recommend service portfolios. The minimum remaining amount of intelligent devices in the network is relatively flat, and the decline trend is slow. Because the life cycle of the network is the time range until the remaining energy of the first intelligent device in the network is insufficient, the initial energy of all intelligent devices in the network is set to be the same in the experiment. In this case, when the remaining energy of a smart device is relatively small, you should try to select other smart devices in the network that can complete the same function to complete this task. This is helpful to balance the energy consumption of the smart device and thus extend the entire network. Life cycle. Experimental results show that compared with GA and ACO algorithms, the PSO algorithm has a better effect in saving the most energy of smart devices under the current experimental parameter settings.

Fig. 12 shows the variance of the remaining energy of the smart devices in the entire network after using GA, ACO, and PSO algorithms for 30 consecutive executions. Generally, variance is used to measure the degree of deviation between random variables and mathematical expectations. In this experiment, the variance represents the smallest remaining balance of smart devices in the entire network. The larger the value of the variance, the more unbalanced the energy consumption of the smart devices in the network is. The experimental results show that compared with GA and ACO algorithms, after using PSO algorithm for service composition recommendation, the variance of the remaining energy in the entire network is smaller. Although the variance of the entire network increases with the number of algorithm executions, PSO The increase in the variance of the algorithm is generally relatively stable, indicating that each intelligent device in the network is load balanced.

The experimental results is of the above three aspects show that in the experimental environment of this paper, the PSO algorithm has better results in the three aspects of the objective function value, the

minimum residual energy and the residual energy variance in the entire network.

5. Conclusion

First, encapsulate the functionality of smart devices in the IoT environment into services, each representing a different function. Functional clustering is divided into service classes according to the degree of semantic similarity of their functions. Each service class represents only a collection of similar functional services, regardless of the non-functional factors of the service. By computing the possibility of invocation between service classes, a service network is constructed after a certain depreciation of pruning. The service network is a directed graph structure that represents a service class connection diagram with certain calling possibilities in the network. Second, build a service class chain based on the user's semantic service request. The depth-first graph search algorithm is used to first determine the initial state and the end state of the service class chain, and find a service recommendation chain in the service network to form a candidate recommendation set, and instantiate the service class chain. Finally, this paper simulates the generation of multiple intelligent devices, using genetic algorithm, ant colony algorithm and particle swarm algorithm to instantiate the service combination, the objective function value, the minimum residual energy of the smart device in the network and the whole network. The device uses load balancing for analysis. The experimental results show that the particle swarm optimization algorithm can obtain a relatively good IoT service combination in the experimental environment of this paper, and is superior to genetic algorithm and ant colony algorithm in performance and energy consumption. There are still many shortcomings in the research of this paper, which need to be further explored and further optimized in future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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